

COTIVITI INTERN ASSESSMENT · TOPIC 2

Agentic AI for Payment Integrity

Why the Auditable Agent Wins

Clinical decision-making and pattern recognition: agentic generative AI for Treatment, Payment, and Operations.

Abdel-Basit Jimoh · Indiana University Bloomington · June 2026

The opportunity hiding in improper payments

\$95.5B

estimated improper payments across Medicare
& Medicaid, FY2025

CMS, 2026

\$28.83B

in Medicare fee-for-service alone

CMS, 2026

Most of it isn't fraud.

It's coding and documentation error:
structured, rule-bound work that an agent
can actually review at volume.

Why this matters for Cotiviti

Payment integrity is pattern recognition at national scale. A first pass that is faster, steadier, and able to explain itself turns a multi-billion-dollar error rate into something a team can actually work through. The open question was never whether AI can read a claim. It's whether it can be trusted to.

Classic methods, plus a new reasoning layer

The familiar toolkit



Classification

Sort a claim into a fixed category: clean or improper.



Prediction

Estimate an unknown value, like what a claim should cost.



Clustering

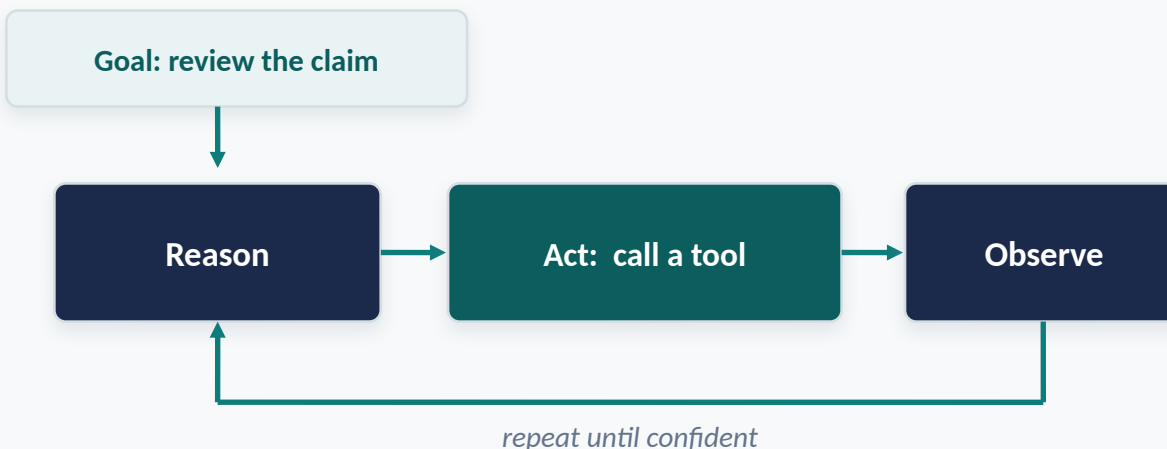
Group providers who bill alike, flag the ones who don't.



Anomaly detection

Watch values over time, catch a sudden billing spike.

THE NEW LAYER: AGENTIC GENERATIVE AI



ReAct in one loop: reason, act, observe, repeat, then return a verdict. Chain-of-thought keeps every step explicit and auditable (Wei 2022; Yao 2023).

Where it sits: across Treatment, Payment, and Operations, Payment is where these methods pay off most clearly and most measurably.

Three shifts converging on payment AI

01

From answers to agents

The field moved from one-shot generative models to agents that take multi-step action under human oversight. Payers are among the first to put it to work, on claims and prior authorization.

02

RAG keeps models honest

Retrieval-augmented generation pulls the real policy text before the model answers, so the output is grounded in source. It's become the default way to cut hallucination in healthcare LLMs.

Amugongo et al., 2025

03

Governance goes operational

Oversight and auditability are turning into design requirements (NIST AI RMF). Newer work tackles “agent sprawl” with identity, kill-switches, and audit logging.

NIST 2023 · Prakash et al., 2026

For a payer, none of this is red tape. It's the part that makes an agent safe enough to actually ship.

Agentic Claim-Review Assistant

The agent takes one synthetic claim, works through the coding and payment rules with three tools, and returns a structured verdict: APPROVE or FLAG, plus a rationale, a confidence score, and a recommendation for the human reviewer.

Streamlit UI: pick a synthetic claim, then hit Review

Agent loop: Groq llama-3.3-70b, native tool-calling, no framework

Picks a tool, reads the result, keeps reasoning until it lands on a verdict.

lookup_policy()

coding/payment rules for the CPT
RAG stand-in

check_eligibility()

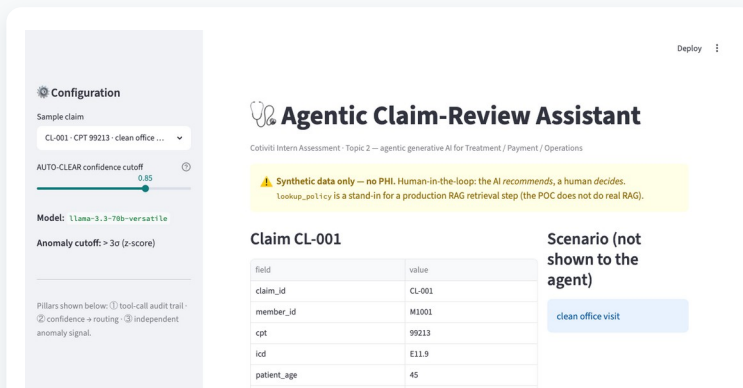
is the member active & covered?

amount_anomaly()

z-score vs. billing history
independent signal

Strict JSON verdict

{ decision, reasons[], confidence, recommended_action }



The running app

Single-file Streamlit. 100% synthetic data, no PHI.
lookup_policy plays the part of a production RAG retrieval step.

Built from first principles, no LangChain or LlamaIndex.

One agent, three routing outcomes

CL-001 • CLEAN OFFICE VISIT

Every rule passes and the amount is unremarkable.

✓ **APPROVE**

Confidence **90%**

AUTO-CLEAR

Confidence sits at or above the 85% cutoff, so it clears with no human needed.

CL-010 • ELEVATED AMOUNT

Passes every hard rule, but the billed amount is unusual ($z \approx 2.0$, just under the 3σ cutoff).

✓ **APPROVE**

Confidence **~75%**

ROUTE TO HUMAN

Approved, but confidence lands below the cutoff, so the threshold alone sends it to a reviewer.

CL-007 • AMOUNT OUTLIER

Passes every hard rule, but the independent anomaly check fires anyway.

🚩 **FLAG**

Confidence **90%**

ROUTE TO HUMAN

Amount anomaly $z = 5.82$ for CPT 27447. A flag always routes to a human, whatever the confidence.

Three design choices, made visible on screen

1

Auditable by construction

Every tool call and its result is a recorded step. That chain of reasoning is the audit trail, the same record a regulator or an appealing provider would ask for.

2

Confidence as a routing threshold

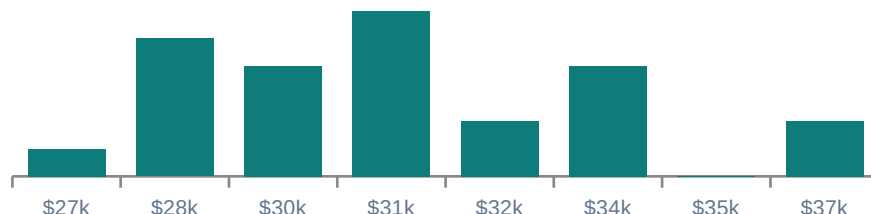
The fix for LLM overconfidence (Tian et al., 2025): calibrate confidence and route on it. Only high-confidence approvals auto-clear; everything else goes to a person.

3

Defense in depth

amount_anomaly is an independent z-score check that runs next to the LLM. The verdict never rests on the model by itself, because a second, classical signal stands behind it.

Pillar ③ in numbers: CL-007, CPT 27447 (n=24)



This claim: \$48,000
History mean \$31,918
 $z = 5.82$ (5.8σ outlier)
Fires even though the hard \$ cap passed.

RAG honesty

The POC skips real RAG to stay simple. lookup_policy plays the part of a production retrieval step: same agent loop, simpler backend. The demo is upfront about that and never fakes retrieval it isn't doing.

A faster, more consistent, explainable first pass



Automation at scale

Pull the policy, check eligibility, reason through whether the diagnosis justifies the procedure, return a decision with a rationale and a confidence score. The structured bulk of review.



Classical methods complement

Clustering and anomaly detection catch providers whose billing breaks from their peers, no labeled data needed. Often the only way a new abuse pattern surfaces (De Meulemeester et al., 2025).



Aligned with Cotiviti's stance

Cotiviti already frames AI as a tool, not a solution: valuable when paired with human expertise and past results to learn from (Arnold & Periyanan, 2025).

This was never about replacing reviewers. It hands them a first pass that's quicker, steadier, and already explained.

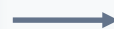
Every risk has a control built into the design

THE RISK



Hallucination

91.8% of clinicians have hit one (Kim 2025)



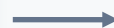
Overconfidence

LLM-as-judge runs hot (Tian 2025)



Bias

A model can relearn yesterday's mistakes



Auditability

Each call must be justified to regulators



THE SAFEGUARD



Tool-grounded retrieval plus visible chain-of-thought; the explicit steps catch errors.



Confidence is calibrated and used as a routing gate, never trusted at face value.



A human signs off on every non-cleared claim, with the full trail to inspect.



The reasoning is the record; every decision traces back to the policy it used.

Make the auditable agent the differentiator

Build agentic claim review as a human-in-the-loop system with a full audit trail, and treat that audit trail as the competitive edge rather than a compliance checkbox.

Defense in depth — each layer is an independent check, with a person at the end:



A claim decision is only as good as the case behind it. Build the agent that shows its work, and the audit trail becomes the pitch.

References

- Amugongo, L. M., et al. (2025). Retrieval augmented generation for LLMs in healthcare: A systematic review. PLOS Digital Health, 4(6).
- Arnold, B., & Periyanan, A. (2025). The evolving role of AI in payment integrity [White paper]. AHIP / Cotiviti.
- Centers for Medicare & Medicaid Services. (2026). Fiscal year 2025 improper payments fact sheet.
- De Meulemeester, H., et al. (2025). Explainable unsupervised anomaly detection for healthcare insurance data. BMC Med. Informatics & Decision Making, 25(1).
- Kim, Y., et al. (2025). Medical hallucinations in foundation models and their impact on healthcare. arXiv:2503.05777.
- National Institute of Standards and Technology. (2023). AI Risk Management Framework (AI RMF 1.0), NIST AI 100-1.
- Prakash, C., Lind, M., & Sisodia, A. (2026). Agentic AI governance and lifecycle management in healthcare. arXiv:2601.15630.
- Tian, Z., et al. (2025). Overconfidence in LLM-as-a-judge: Diagnosis and confidence-driven solution. arXiv:2508.06225.
- Wei, J., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. NeurIPS, 35.
- Yao, S., et al. (2023). ReAct: Synergizing reasoning and acting in language models. ICLR.